

Isoflurane: Drug information - UpToDate

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For additional information see "Isoflurane: Patient drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Forane;
- Terrell

Pharmacologic Category

- General Anesthetic, Inhalation

Dosing: Adult

Note: Coughing, breath-holding, bronchospasm, or laryngospasm may occur during induction; use of an ultra short-acting barbiturate may avoid these symptoms. Minimum alveolar concentration of isoflurane decreases with increasing patient age. Select premedication according to the need of the individual patient, taking into account that secretions are weakly stimulated by isoflurane, and the heart rate tends to be increased. Dosage must be individualized based on patient response.

Anesthesia

Anesthesia: Inhalation:

Induction: 1.5% to 3% isoflurane with oxygen or oxygen-nitrous oxide mixture.

Maintenance: With nitrous oxide: 1% to 2.5%; additional 0.5% to 1% with oxygen alone.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, dosage adjustment unlikely as limited systemic absorption is expected.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Dosing: Older Adult

Refer to adult dosing. Minimum alveolar concentration is decreased in older patients.

Dosing: Pediatric

Anesthesia

Anesthesia: Inhalation: Refer to adult dosing.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported in adult and pediatric patients.

>10%:

Gastrointestinal: Nausea (recovery: 15%)

Nervous system: Agitation (induction: 52%), chills ($\leq 14\%$), shivering ($\leq 14\%$)

Respiratory: Breath-holding (induction: 24%; maintenance: 1%), cough (induction: 28%; maintenance: 4%)

1% to 10%:

Cardiovascular: Atrial arrhythmia (intraoperative: 2%), atrioventricular nodal arrhythmia (intraoperative: 2% to 4%), cardiac arrhythmia (postoperative: 1%), ventricular arrhythmia (intraoperative: 2% to 3%)

Gastrointestinal: Retching (induction, maintenance: $\leq 1\%$), vomiting (recovery: 10%; induction: $< 1\%$)

Nervous system: Delirium (6%), involuntary muscle movements (maintenance: 2%)

Neuromuscular & skeletal: Laryngospasm (induction: 8%; maintenance: $< 1\%$)

$< 1\%$:

Cardiovascular: Hypertension (intraoperative, postoperative), hypotension (intraoperative, postoperative)

Dermatologic: Diaphoresis (induction)

Nervous system: Abnormal electroencephalogram (convulsive pattern), mood changes, nightmares, seizure

Respiratory: Change in bronchial secretions (induction)

Frequency not defined:

Endocrine & metabolic: Decreased serum cholesterol, increased lactate dehydrogenase, increased serum glucose

Hematologic & oncologic: Leukocytosis

Hepatic: Abnormal hepatic function tests (bromsulphthalein clearance decreased), decreased serum alkaline phosphatase, increased serum alanine aminotransferase, increased serum alkaline phosphatase, increased serum aspartate aminotransferase, increased serum bilirubin

Nervous system: Asthenia, ataxia, cognitive dysfunction, confusion, dizziness, drowsiness, fatigue, nervousness

Neuromuscular & skeletal: Myalgia

Renal: Decreased blood urea nitrogen, increased serum creatinine

Postmarketing:

Cardiovascular: Acute myocardial infarction, atrial fibrillation, bradycardia, complete atrioventricular block, first-degree atrioventricular block, flushing, ischemic heart disease, low cardiac output, premature ventricular contractions, prolonged QT interval on ECG, second-degree atrioventricular block, tachycardia, torsades de pointes, ventricular fibrillation, ventricular tachycardia

Dermatologic: Skin rash

Endocrine & metabolic: Hyperkalemia (with underlying myopathies)

Gastrointestinal: Hiccups, pancreatitis

Genitourinary: Oliguria

Hematologic & oncologic: Increased hemoglobin (carboxyhemoglobin)

Hepatic: Cholestatic hepatitis, fulminant hepatitis, hepatic failure, hepatic necrosis, hepatitis, hepatotoxicity (idiosyncratic; Chalasani 2021), increased gamma-glutamyl transferase, jaundice, liver steatosis

Hypersensitivity: Hypersensitivity reaction (including anaphylaxis)

Nervous system: Awareness under anesthesia without pain, brain edema, headache, hypothermia, increased intracranial pressure, migraine, myoclonus, withdrawal syndrome (following multi-day exposure)

Neuromuscular & skeletal: Rhabdomyolysis

Ophthalmic: Anisocoria, nystagmus disorder

Renal: Acute kidney injury

Respiratory: Airway obstruction, apnea, bronchospasm, hypercapnia, hypoxia, respiratory depression, stridor

Miscellaneous: Malignant hyperthermia

Contraindications

Known sensitivity to isoflurane, other halogenated agents, or any component of the formulation; patients in whom general anesthesia is contraindicated; known or suspected genetic susceptibility to malignant hyperthermia; history of hepatitis due to a halogenated inhalational anesthetic or a history of unexplained moderate to severe liver dysfunction (eg, jaundice associated with fever and/or eosinophilia) after anesthesia with isoflurane or other halogenated inhalational anesthetics.

Warnings/Precautions

Concerns related to adverse effects:

- Cardiovascular effects: Decrease in BP is dose dependent due to peripheral vasodilation; cardiac output is maintained. Use caution in patients who are hypovolemic, hypotensive, or hemodynamically compromised; use caution in patients with coronary artery disease to avoid risk of myocardial ischemia. May prolong the QT interval and rarely torsades de pointes; use caution in patients at risk of QT prolongation. May produce cardiac steal (due to coronary vasodilation) and reflex tachycardia, but does not depress cardiac conduction nor does it sensitize the myocardium to catecholamine-induced arrhythmias like halothane (not available in the United States) (Golembiewski 2004).
- CNS depression: Advise patients to wait at least 24 hours after administration before engaging in activities which require mental alertness (operating machinery or driving).
- Decreased blood flow: May cause decrease in hepatic, renal, and splenic blood flow (Gelman 1984).
- Hepatic effects: Postoperative mild to severe hepatic dysfunction (including fatal hepatic necrosis and hepatic failure) and hepatitis with or without jaundice have been reported; may cause sensitivity hepatitis in patients who have been sensitized by previous exposure to halogenated anesthetics. Risk may be increased with repeated isoflurane exposure in a short time.

- **Hyperkalemia:** Use of inhaled anesthetics has been associated with rare cases of perioperative hyperkalemia that have resulted in cardiac arrhythmias (including fatalities) in pediatric patients; concomitant use of succinylcholine was associated with many of the reported cases, but not all. Risk of hyperkalemia is increased in pediatric patients with underlying neuromuscular disease (eg, Duchenne muscular dystrophy). Other abnormalities may include elevation in creatinine kinase (CK) and myoglobinuria. Aggressively identify and treat the hyperkalemia and resistant arrhythmias.
- **Hypersensitivity:** Hypersensitivity reactions (eg, anaphylaxis, cardiovascular collapse, difficulty breathing, hypotension, rash) have been reported.
- **Increased intracranial pressure:** Dilates the cerebral vasculature and may, in certain conditions, increase intracranial pressure (ICP). Administer in conjunction with ICP-reducing strategies, as clinically needed.
- **Malignant hyperthermia:** May trigger malignant hyperthermia; some reported cases have been fatal. Risk may be increased with concomitant administration of succinylcholine and volatile anesthetic agents and patients with genetic factors or family history of malignant hyperthermia, including ryanodine receptor or dihydropyridine receptor inherited variants. Signs of malignant hyperthermia may include arrhythmias, cyanosis, hemodynamic instability, hypercapnia, hyperthermia, hypovolemia, hypoxia, muscle rigidity, tachycardia, and tachypnea; coagulopathies, renal failure, and skin mottling may also occur. If malignant hyperthermia is suspected, discontinue triggering agents and institute appropriate therapy (eg, dantrolene) and other supportive measures.
- **Obstetrical anesthesia:** Increased blood loss comparable with that seen with halothane has been reported during abortions.
- **Respiratory depression:** Causes dose-dependent respiratory depression and blunted ventilatory response to hypoxia and hypercapnia (Golembiewski 2004).

Disease-related concerns:

- **Heart failure:** In a scientific statement from the American Heart Association, isoflurane has been determined to be an agent that may exacerbate underlying myocardial dysfunction (magnitude: major) (AHA [Page 2016]).
- **Hepatic impairment:** Use with caution in patients with history of hepatic impairment or taking medications known to cause hepatic impairment; evaluate need for repeated exposure in a short time as clinically indicated.

Special populations:

- **Pediatric neurotoxicity:** In pediatric and neonatal patients <3 years and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems including neurodevelopmental delay (and related diagnoses), learning disabilities, and ADHD. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed.

Special handling:

- **Occupational caution:** There is no specific work exposure limit established for isoflurane. However, the National Institute for Occupational Safety and Health (NIOSH) recommends no worker be exposed to >2 ppm (ceiling concentrations) over a period of 1 hour. Precautions (eg, adequate ventilation, scavenging-systems, minimizing leaks/spills) can help to lessen any potential risk.

Other warnings/precautions:

- **Desiccated absorbents:** Reaction of isoflurane with desiccated CO₂ absorbents produce carbon monoxide, which may result in elevated carboxyhemoglobin levels in some patients. Replace any

desiccated CO₂ absorbents prior to administration; the color indicator of most CO₂ absorbents may not change when it becomes desiccated and, therefore, replacement of CO₂ absorbents routinely regardless of color change is recommended.

- **Monitoring:** Transient elevations in burst suppression pattern retention, blood glucose, and SCr with decreases in alkaline phosphatase, BUN, and serum cholesterol may occur.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Inhalation:

Forane: USP: ≥ 99.9 mL/100 mL (100 mL, 250 mL)

Terrell: USP: ≥ 99.9 mL/100 mL (100 mL, 250 mL)

Generic: USP: ≥ 99.9 mL/100 mL (100 mL, 250 mL)

Generic Equivalent Available: US

Yes

Administration: Adult

Via isoflurane-specific calibrated vaporizer or vaporizer with calculated flow required

Administration: Pediatric

Inhalation: Administer via isoflurane-specific calibrated vaporizer or vaporizer with calculated flow required

Storage/Stability

Store at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Anesthesia: For induction and maintenance of general anesthesia. Adequate data have not been developed to establish its application in obstetrical anesthesia.

Medication Safety Issues

Sound-alike/look-alike issues:

Isoflurane may be confused with enflurane

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

Substrate of CYP2E1 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acrivastine: May increase adverse/toxic effects of Inhalational Anesthetics. *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amisulpride (Oral): May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk C: Monitor*

Amphetamines: May increase hypotensive effects of Inhalational Anesthetics. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bambuterol: May increase arrhythmogenic effects of Inhalational Anesthetics. Management: Some labels recommend specifically avoiding halothane; others recommend separating administration by at least 6 hours; other bambuterol labels do not mention this possible interaction. Monitor for increased sensitivity to arrhythmias if coadministered. *Risk D: Consider Therapy Modification*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Calcium Channel Blockers: Inhalational Anesthetics may increase hypotensive effects of Calcium Channel Blockers. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Dabrafenib: QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dexmethylphenidate-Methylphenidate: May increase hypertensive effects of Inhalational Anesthetics. Management: Avoid the use of dexmethylphenidate/methylphenidate on the day of a planned surgery with an inhalational anesthetic. *Risk D: Consider Therapy Modification*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Domperidone: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

DOPamine: Inhalational Anesthetics may increase arrhythmogenic effects of DOPamine. *Risk C: Monitor*

Doxapram: May increase adverse/toxic effects of Inhalational Anesthetics. Management: Delay administration of doxapram until the inhalational anesthetic has been excreted (ie, after 5 half-lives)

in order to lessen the potential for arrhythmias, including ventricular tachycardia and ventricular fibrillation. *Risk D: Consider Therapy Modification*

DroPERidol: May increase CNS depressant effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). DroPERidol may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Consider dose reductions and monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Ephedra: May increase arrhythmogenic effects of Inhalational Anesthetics. *Risk C: Monitor*

EPHEDrine (Nasal): May increase arrhythmogenic effects of Inhalational Anesthetics. *Risk X: Avoid*

EPINEPHrine (Systemic): Inhalational Anesthetics may increase arrhythmogenic effects of EPINEPHrine (Systemic). Management: Administer epinephrine with added caution in patients receiving, or who have recently received, inhalational anesthetics. Use lower than normal doses of epinephrine and monitor for the development of cardiac arrhythmias. *Risk D: Consider Therapy Modification*

Esketamine (Injection): May increase sedative effects of Inhalational Anesthetics. *Risk C: Monitor*

Fenfluramine: May increase hypotensive effects of Inhalational Anesthetics. *Risk C: Monitor*

Fenoterol: Inhalational Anesthetics may increase arrhythmogenic effects of Fenoterol. *Risk C: Monitor*

Fluorouracil Products: QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Haloperidol: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Isocarboxazid: May increase hypotensive effects of Inhalational Anesthetics. *Risk X: Avoid*

Isoproterenol: Inhalational Anesthetics may increase arrhythmogenic effects of Isoproterenol. *Risk X: Avoid*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levoketoconazole: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Metaraminol: Inhalational Anesthetics may increase arrhythmogenic effects of Metaraminol. *Risk X: Avoid*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methadone: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase CNS depressant effects of Methadone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation, ventricular arrhythmias, sedation, and respiratory depression. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Neuromuscular-Blocking Agents (Nondepolarizing): Inhalational Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Management: When initiating a non-depolarizing neuromuscular blocking agent (NMBA) in a patient receiving an inhalational anesthetic, initial NMBA doses should be reduced 15% to 25% and doses of continuous infusions should be reduced 30% to 60%. *Risk D: Consider Therapy Modification*

Neuromuscular-Blocking Agents: Isoflurane may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nitrous Oxide: May increase therapeutic effects of Isoflurane. Specifically, isoflurane MAC values may be decreased. Isoflurane may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Norepinephrine: Inhalational Anesthetics may increase arrhythmogenic effects of Norepinephrine. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Ondansetron: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentamidine (Systemic): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Phenylephrine (Ophthalmic): May increase adverse/toxic effects of Inhalational Anesthetics. Specifically, the cardiovascular depressant effects of inhalational anesthetics may be increased. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pimozide: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Piperaquine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pyridostigmine: May increase or decrease neuromuscular-blocking effects of Inhalational Anesthetics. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Antidepressants (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Antipsychotics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Antipsychotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Class IC Antiarrhythmics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): May increase hypotensive effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for hypotension and QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Miscellaneous Agents (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Reproterol: Inhalational Anesthetics may increase arrhythmogenic effects of Reproterol. *Risk C: Monitor*

Ritodrine: May increase adverse/toxic effects of Inhalational Anesthetics. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Thioridazine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Pregnancy Considerations

Isoflurane crosses the placenta (Dwyer 1995; Satoh 1995).

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D- aspartate (NMDA) receptors and/or potentiate gamma-aminobutyric acid (GABA) activity, may affect brain development. Evaluate benefits and potential risks of fetal exposure to isoflurane when duration of surgery is expected to be >3 hours (Olutoye 2018).

Use of isoflurane in obstetric anesthesia has been described (Abboud 1989; ACOG 209 2019; Devroe 2015; Ghaly 1988; Liu 2013). Maternal exposure should be minimized due to dose dependent uterine relaxation and fetal depression (ACOG 209 2019; Devroe 2015).

The ACOG recommends that pregnant patients should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

It is not known if isoflurane is present in breast milk.

According to the manufacturer, the decision to breastfeed should consider the risk of infant exposure and the benefits of breastfeeding to the infant. Based on pharmacokinetic properties, use of isoflurane in breastfeeding patients may be considered acceptable (Lee 1993). The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Burst suppression pattern, BP, electrocardiogram, serum potassium, SCr, BUN, oxygen saturation, respiration, end-tidal CO₂ and isoflurane concentrations should be monitored prior to and throughout anesthesia; consider additional cardiac monitoring in patients with coronary artery disease; QT interval in patients at risk for QTc prolongation; blood glucose; alkaline phosphatase; serum cholesterol.

Mechanism of Action

Isoflurane is a general anesthetic agent capable of producing profound respiratory depression. Inhaled anesthetics alter activity of neuronal ion channels, particularly the fast synaptic neurotransmitter receptors (nicotinic acetylcholine, gamma-aminobutyric acid [GABA], and glutamate receptors). May depress myocardial contractility, decrease blood pressure through a decrease in systemic vascular resistance, and decrease sympathetic nervous activity (Stachnik 2006).

Pharmacokinetics (Adult Data Unless Noted)

Metabolism: Minimally hepatic (<0.2%), predominantly CYP2E1 (Golembiewski 2004)

Excretion: Exhaled gases

Patient Counseling Points

(For additional information see "Isoflurane: Patient drug information")

What is this drug used for?

- It is used for anesthesia during some procedures or surgery.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Upset stomach or throwing up
- Shivering

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High potassium levels like a heartbeat that does not feel normal; feeling confused; feeling weak, lightheaded, or dizzy; feeling like passing out; numbness or tingling; or shortness of breath
- Dizziness or passing out
- Cough
- Feeling agitated
- Feeling confused
- Trouble breathing, slow breathing, or shallow breathing
- Malignant hyperthermia like a fast heartbeat, fast breathing, fever, or spasm or stiffness of the jaw muscles
- Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Abnormal heartbeat (prolonged QT interval) like a fast or abnormal heartbeat, or if you pass out
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to

provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Aerrane | Forane;
- (AR) Argentina: Forane | Isoflurano | Zuflax;
- (AT) Austria: Forane | Isofluran | Isofluran Rhodia;
- (AU) Australia: Aerrane | Forthane;
- (BD) Bangladesh: Forane;
- (BE) Belgium: Aerrane | Forene | Sedaconda;
- (BG) Bulgaria: Forane | Terrell;
- (BR) Brazil: Isoflurano | Isoforine | Isoran | Isothane;
- (CN) China: Aerrane | Forane | Ning fen | Terrell | Yi mei ning;
- (CO) Colombia: Isoflurano | Isorane | Terrell;
- (CR) Costa Rica: Isoflurano;
- (CZ) Czech Republic: Aerrane | Forane | Isofluran nicholas piramal;
- (DE) Germany: Forane | Forene | Isofluran | Isofluran DeltaSelect;
- (DO) Dominican Republic: Isoforine;
- (EE) Estonia: Aerrane | Forane | Isofluran baxter | Isofluran piramal;
- (EG) Egypt: Aerrane | Anahal | Forane | Isoflurane Sedico | Mira isoflurane;
- (ES) Spain: Aerrane | Forane | Isoflurano inibsa | Sedaconda;
- (ET) Ethiopia: Forane;
- (FI) Finland: Forene | Isofluran baxter | Isofluran Rhodia;
- (FR) France: Aerrane | Cedaconda | Forane | Forene | Forene ecofil;
- (GB) United Kingdom: Aerrane | Forane | Isoflurane abbott | Isoflurane bioglan;
- (GR) Greece: Forenium;
- (GT) Guatemala: Isoflurano | Isotroy;
- (HK) Hong Kong: Aerrane | Forane;
- (HU) Hungary: Aerrane | Forane;
- (ID) Indonesia: Aerrane | Forane | Terrell;
- (IE) Ireland: Forane;
- (IN) India: Aerrane | Forane | Iflurane | Isifrane | Isorane | Isotroy;
- (IT) Italy: Aerrane | Forane;
- (JO) Jordan: Aerrane | Floran | Forane;
- (JP) Japan: Escain | Forane | Isoflurane merck hoei;
- (KE) Kenya: Forane | Isotroy;
- (KR) Korea, Republic of: Aerane | Forane | I-Fran | Rhodiaisoflurane | Terrel;
- (KW) Kuwait: Aerrane | Forane;
- (LB) Lebanon: Forane;
- (LT) Lithuania: Forane;
- (LU) Luxembourg: Forene;
- (LV) Latvia: Forane;
- (MX) Mexico: Dorin | Fluduk | Forane | Isoflurano | Lisorane | Sofloran;
- (MY) Malaysia: Aerrane | Forane | Isorane;
- (NG) Nigeria: Forane | Isotroy;
- (NL) Netherlands: Forene | Isofluraan | Sedaconda;
- (NO) Norway: Aerrane | Forene | Isofluran | Sedaconda;
- (NZ) New Zealand: Aerrane | Forane;
- (PA) Panama: Isoflurano usp | Terrell;

- (PE) Peru: Isoflurano;
- (PH) Philippines: Aerrane | Florane | Forane | Isorane | Pascual isoflurane;
- (PK) Pakistan: Forane;
- (PL) Poland: Aerrane | Forane;
- (PR) Puerto Rico: Forane;
- (PT) Portugal: Aerrane | Forene | Isoflurano;
- (PY) Paraguay: Forane | Isoflurano interlabo | Isoflurano omicron | Isoflurano prosalud | Isoflurano quimfa | Isoflurano usp interlabo | Isoforine | Zuflax;
- (QA) Qatar: Aerrane | Floran;
- (RO) Romania: Aerrane | Anesteran | Forane;
- (RU) Russian Federation: Aerrane | Forane;
- (SA) Saudi Arabia: Aerrane | Floran | Forane;
- (SE) Sweden: Aerrane | Forene | Isofluran baxter | Isofluran Rhodia | Isoflurane Medeva Europe | Sedaconda;
- (SG) Singapore: Aerrane | Forane;
- (SI) Slovenia: Forane | Isofluran;
- (SK) Slovakia: Forane | Isofluran Rhodia;
- (SR) Suriname: Isofluraan;
- (SV) El Salvador: Sofloran;
- (TH) Thailand: Aerrane | Forane | Isofluran Rhodia | Terrell;
- (TN) Tunisia: Forene;
- (TR) Turkey: Aerrane | Forane | Isoflurane usp;
- (TW) Taiwan: Aerrane | Forane;
- (UA) Ukraine: Forane;
- (UG) Uganda: Forane | Isotroy;
- (UY) Uruguay: Forane | Isoflurano Zeneca | Isoforine | Terrell;
- (VE) Venezuela, Bolivarian Republic of: Forene | Isoflurano | Isoflurano usp;
- (ZA) South Africa: Aerrane | Forane | Isofor;
- (ZM) Zambia: Forane | Isotroy;
- (ZW) Zimbabwe: Forane

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